Universiti Kebangsaan Malaysia
The National University of Malaysia

UNIVERSITI WATAN KITA

FACULTY OF SCIENCE & TECHNOLOGY

KELAB STATISTIK UKM

DATA CHALLENGE
UKM STATISTICS CLUB

8 DECENT WORK AND ECONOMIC GROWTH

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



THE ECONOMIC RIPPLE EFFECT OF FOOD WASTE



UKM DATA CHALLENGE 5.0
PRESENTED BY KAW KAW PENGUINS

INTRODUCTION

1. **4.2 BILLION KG FOOD WASTE GENERATED ANNUALLY (2024)**
2. **34.1 MILLION POPULATION**
3. **AVERAGE HOUSEHOLD INCOME OF RM126000**
4. **FOOD PRICES RISING STEADILY**
5. **CPI ↑ INFLATION ↑**

PROBLEM STATEMENT

- Rising food prices
- High food expenditure
- Same level of food waste
- Money is being thrown away in the form of food

SDG ALIGNMENT

SDG 12.3 (Responsible Consumption)

- Half per capita food waste by 2030

SDG 8 (Economic Growth)

- Food waste reduction brings to hidden income boost

OBJECTIVE

QUANTIFY

The financial value of food waste by state
"How much money is in the bin?”

MAP

Relationship between Income levels and Food expenditure
"What is the proportion income spent on food?”

ANALYZE

How CPI and Inflation drive food waste behaviour
"Do higher prices force smarter consumption in people?”

FORECAST

10-year projections for:

- Food waste
- Food expenditure
- CPI

"Where are we heading if nothing changes?”

METHODOLOGY

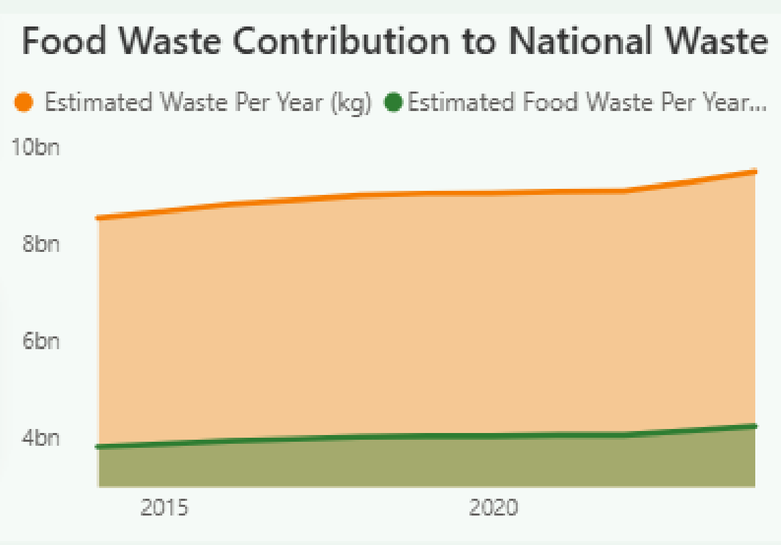
Data Layer	Source / Method	Purpose
Food Waste	State-level estimates (kg/year)	Identify high and low waste area
Income	Household survey data	Calculate food expenditure ratio
Expenditure	% of income spent on food	Measure economic burden
CPI	Yearly index	Track price pressure over time
Inflation	Year-over-year % change	Understand purchasing power
Forecasts	Time series modeling	Project 10-year trends

EXCEL

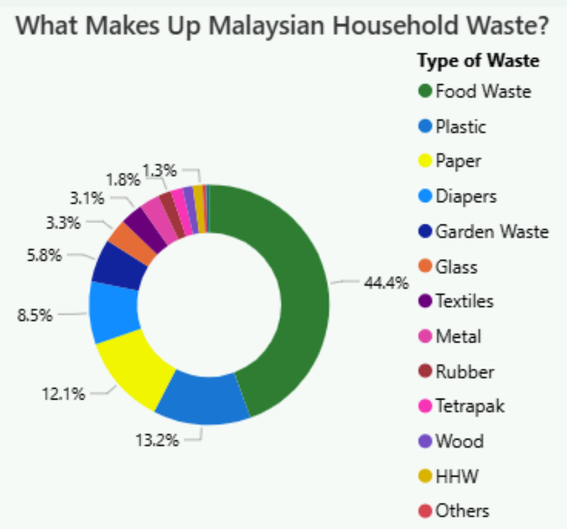
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POWER BI

MALAYSIA'S GROWING FOOD WASTE CRISIS

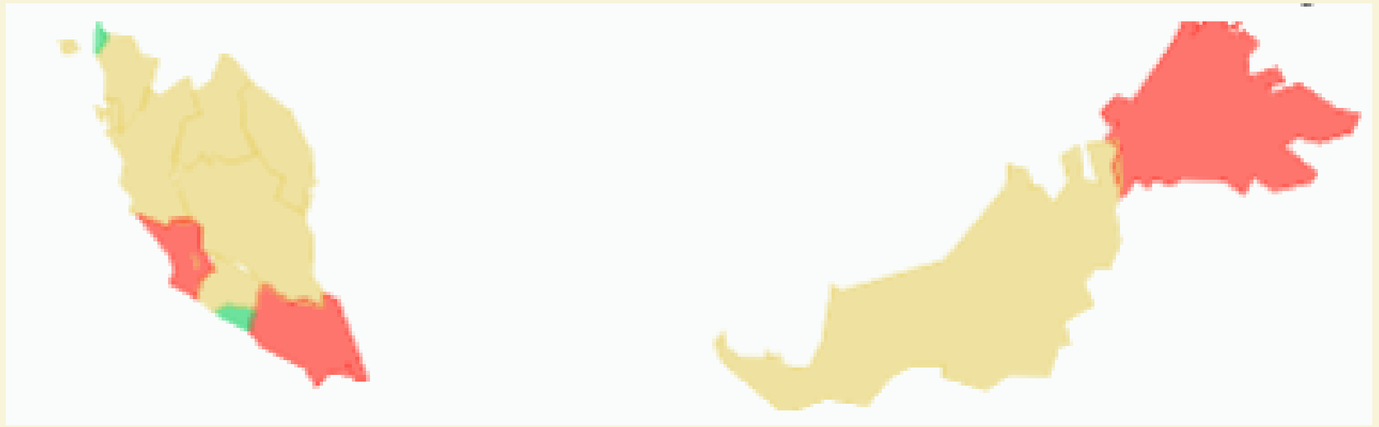


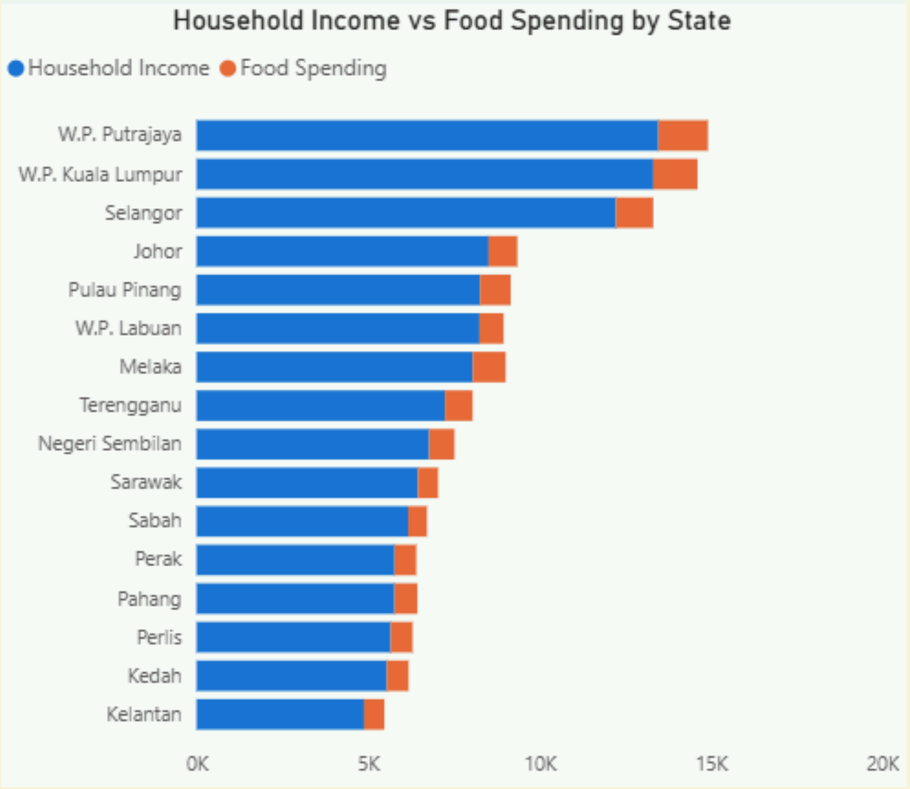
- A rising trend of food waste discarded yearly
- increasing in an alarming rate in recent years



- Food waste makes up roughly 44% of Malaysia’s total municipal solid waste.

WHO ARE THE LEAD CONTRIBUTOR?



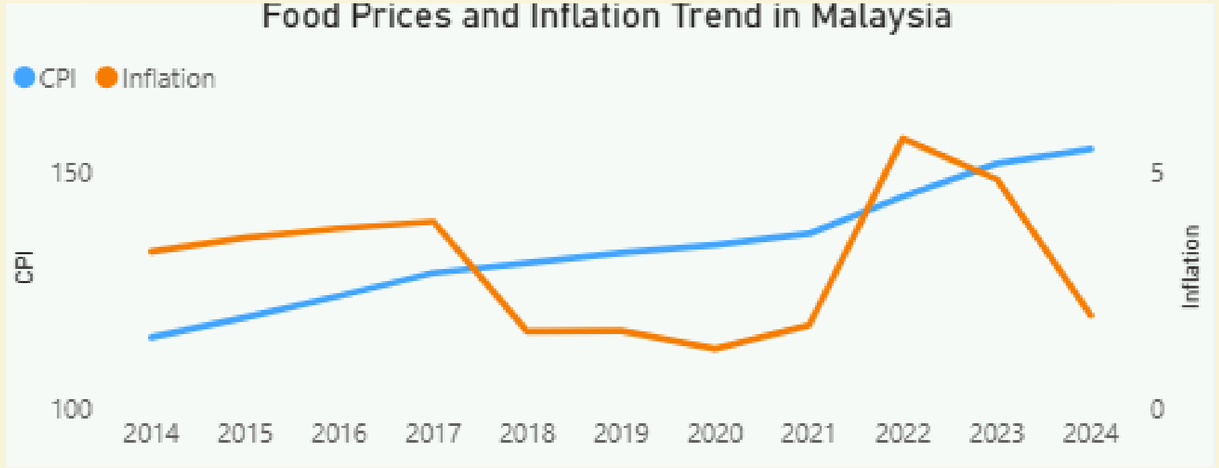


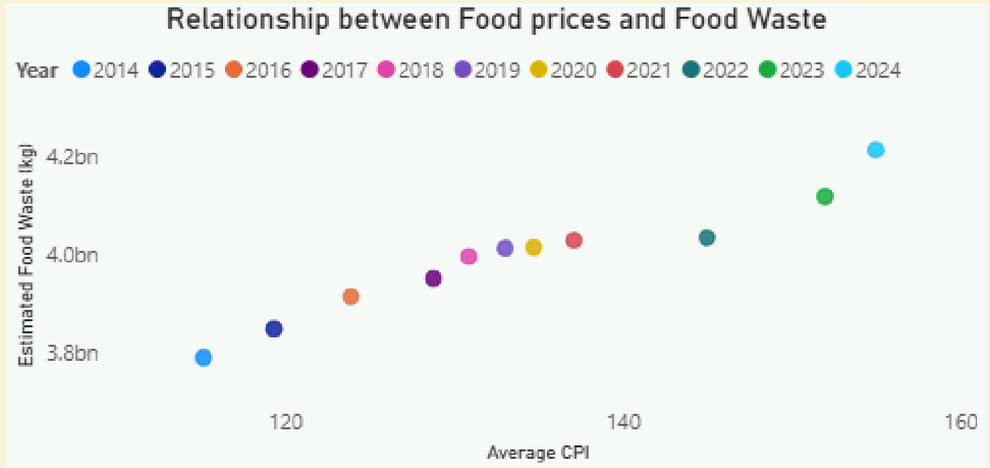
HIGHEST FOOD WASTE : SELANGOR, JOHOR, SABAH

LOWEST FOOD WASTE : PERLIS, PUTRAJAYA, LABUAN

- Population of states influence amount of waste generated
- Average household income is RM126000
- Economic growth in households causes more money spent on food

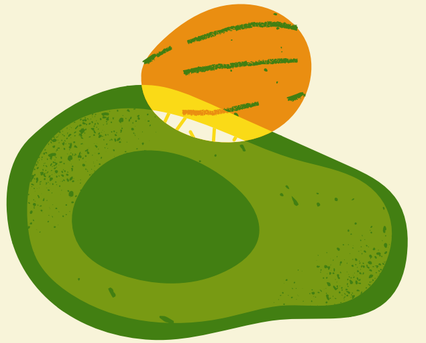
THE INFLATION-WASTE PARADOX



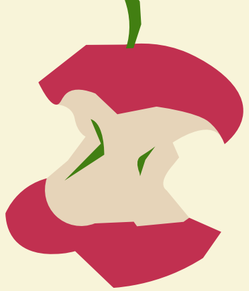


- Even though food inflation and CPI has been a major concern, food waste volumes remain high.
- An average family could be wasting RM200+ per month simply through uneaten food
- Higher food prices do not necessarily reduce food waste





FOOD WASTE CALCULATOR



Step 1 : Select Your Monthly Food Spending

Monthly Food Spending for each household

985

Step 2 : Select Your Monthly Food Waste

Food Waste per Month (kg)

30

Step 3 : Select Your Waste Reduction Goal

Food Waste Reduction Goal (%)

0.50

Monthly Food Waste Cost

RM134.32

Annual Food Waste Cost

RM1.61K

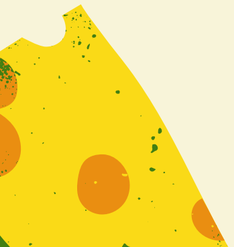
Potential Monthly Savings

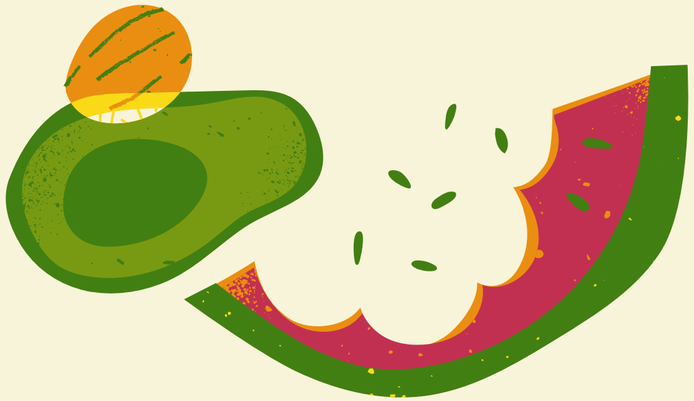
RM67.16

Potential Annual Savings

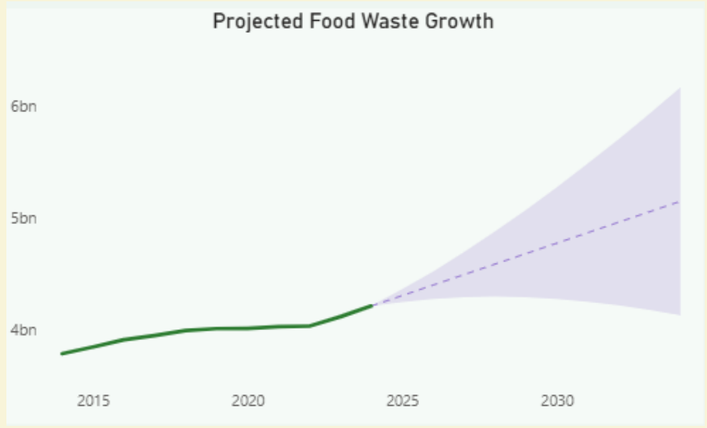
RM805.91

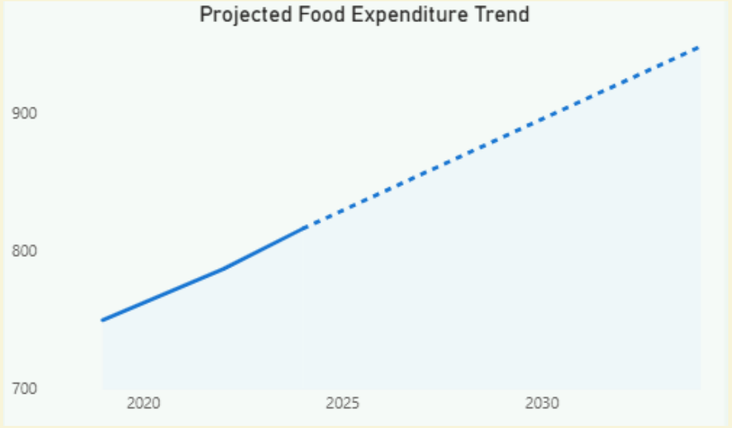
- The calculator allows users to input their own spending, food waste and their goal
- Allows users to see how much they could save
- While we see waste as "trash," this tool redefines it as "lost savings"

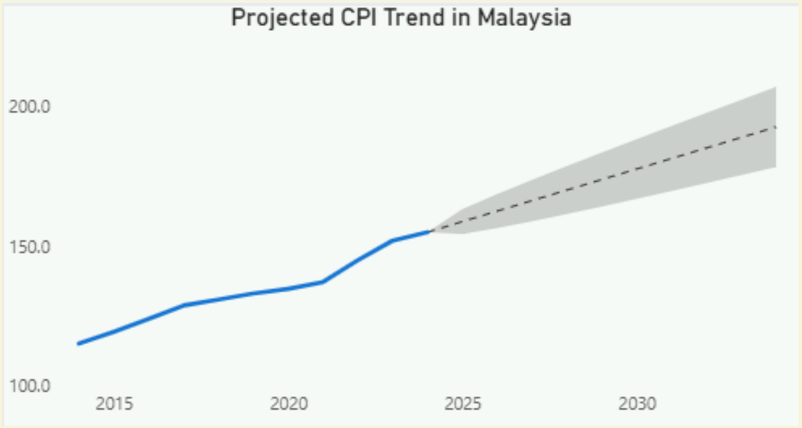




FUTURE & FORECAST







IN 2034 :

- 5.1 billion kg of food waste generated
- RM 948 spent on food monthly
- Consumer price index increases by 37.67 units in 10 years



CONCLUSION AND RECOMMENDATION

SIZE	INEQUALITY	BEHAVIOR	FUTURE
4.2 Billion kg of waste in 2024 = Billions of ringgit loss	• Household spends 35% of their income on food • inflation does not reduce waste	• exceed expiration date • spoilt food • dislike in food	5.1 Billion kg of food waste with no interventions



- SMART SHOPPING LISTS
- ENCOURAGES 5-10% "INCOME BOOST"



- TARGET TOP WASTE STATES
- TRACK FOOD WASTE AS AN ECONOMIC INDICATOR (NOT JUST ENVIRONMENTAL)



- INTEGRATE WASTE TRACKING INTO HOUSEHOLD SURVEYS
- CREATE DASHBOARDS FOR PUBLIC AWARENESS


TOGETHER FOR
A WASTE-FREE
FUTURE